Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · CLASS 41.1 Beams Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 41.1 Beams Numerical

1 CLASS 41.1 Beams Numerical

Study focus: CLASS 41.1 Beams_Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000118 · EV0000119

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is the first question for today's class.

Key details

- Determine the reactions at A and B for the beam loaded as shown.
- There is a beam AB.
- What's the name of this beam?
- Do you remember?
- This part of the beam, this 4 meter is extended behind the support, hence it is called as woven hanging beam.
- It is also called as combination because it's a combination of pin, support, and roller reaction.
- A beam sorry This load 18 kilon per meter it is a UDL spread it over a span of four meters.
- Couple 15 kilon meter and it's a free vector.
- Remember this it's a free vector, even though they have specified some point like two meter here with respect to N with respect to B.
- It doesn't matter where it acts because it's a free vector.
- The moment of a couple is a free vector.
- The magnitude is 15 kilon meter.
- Now the first is to find the support reactions AX AY then reaction at B.
- So the first thing is we have to start with the free body diagram.
- See, before that we have to convert this UDL into a point load in the free body diagram.
- How to convert UDL into a point load?
- We know the formula, the resultant of the UDLR will be equal to.
- W into L A F rectangle.

- What is W here eighteen kilonewton per meter and what is L span over which UDL is acting four meter.
- The various 72 kilo newton.
- Next, where exactly the 72 kilo newton will be acting?
- Midpoint, nothing but with respect to B, L by 2, and with respect to the free end, L by 2.
- Remember we have to project all these points vertically downward, then draw a horizontal line, this will be the beam AB.
- Next start from any one of the end, either from A or from this free end.
- I am going to start with the end A.
- Since it is a pin connection, two reactions, one along X direction, one more Y direction.
- Next moving this way.
- Look at this, you have to move in this direction.
- Then after two meter, the mentioned the moment of a couple fifteen kilonewton meter, take that as it is.
- Next, after two meter we don't have any loading here.
- Then at B it's a roller support.
- that will be having only one reaction normal to the surface of contact so RB R B will be vertically upward.
- That way exactly that will be acting.
- As I said before, midpoint, L by 2 then L by 2.
- Yes, we have considered each and every force acting on the beam.
- How to find AXA by R B?
- Since it is a general force system, we have to apply the three conditions of equilibrium, sigma fx, 0, f y 0, then momentum about any point, 0.
- That's it.
- Next y direction we have two react two unknowns A by then R B.
- Before that I'll apply the moment equation.
- Moment about AS 0.
- This is the second condition of equilibrium.
- Look at this A.
- At A we have two reactions, but moment of these two will be zero.
- Then moving the.
- There is a moment of a couple 15 kN meter.
- Look at the direction, is it clockwise or counterclockwise?
- It is clockwise, so it should be taken as minus 15.
- And do we need to multiply the distance?
- Since it's kilonewtan meter, already the distance is there, it's a free vector.
- Don't multiply the distance per moment of a couple.
- Minus 15, we are done with the moment of this.
- Then come to this RB, RB is in the counterclockwise direction.
- Apply the right hand top rule.
- R B is in anti clockwise direction.

Example or application from the lecture

- First, let's look into this beam.
- Let's start with a free body diagram of the beam.
- Next four meters we have a UDL we have to consider the result end of that UDL that is 72 kilonewton.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 05:00–09:59

Next, which is an X load 72 kilonton.

Key details

- It is acting in a clockwise direction like this.
- Clockwise with respect to A.
- What is the perpendicular distance from this point to A?
- It is 2 plus 2.
- 4 plus 2.
- There is only one unknown in this equation that is RP.
- If you simplify you'll get the value of R B as positive.
- 111.75 kilo Newton.
- This plus sign indicates that assumption is correct.
- We assume the RB is up, but force that's correct.
- Next we are left with only one more unknown that is AY.
- Apply the sigma FY 0.
- So next is the RB plus R B because it is upward then minus 72 is equal to 0.
- Since we know the value of RB, substitute RB, you will get the value of Ay.
- Please make sure so AY is actually acting in the downward direction.
- Determine the reactions at A, B, and D for the pair of beams connected by ideal pin at C and subjected to the concentrated and distributed loads.
- Look at this figure.
- There are two beams A, C, and C D.
- These two are connected by IDL pin at C.
- If you think of finding the support reactions A, C, Y, R B, then.
- R D by considering beam A and D, we have more number of support reactions, then it will become statically indeterminate because we can't find all the reactions using the conditions of equilibrium.
- Then what we can do here is.
- We can divide this into two games, A C then C D.
- Here two reactions at A, then one reaction at B.
- Today we'll have like five support reactions when we talk about being AC in usually.
- Next option is being C D.
- If you think of C D, so two reactions at C, then.
- Only one reaction at D.
- This is how you have to solve this particular question.

- First one is considered being C D.
- Before that we have to find the resultant of this U V L in your point.
- What is the formula to find the resultant of UVL?
- We know the formula R is equals to half into base into height.
- Half into base into height.
- What is height here?
- 5 kilon meter.
- Then what is the base here span over which the load is acting?
- Next, where exactly this 10 will be acting from this end, nothing but from C, it will be at a distance of L by 3, or from this end it will be at a distance of 2 L by 3.
- First, project all the points vertically downward, draw a horizontal line, this will be the beam C D.
- Then start from any one of the end.
- I am going to start from this end, C.
- At C it's a pin.
- Next we have move in this direction towards right, we have UVL.
- What is the results of UV Lenkil Newton?
- Where exactly is that it is acting?
- L by 3 with respect to C.
- Resultant is 10 kilometers.
- Then what is the remaining distance from year to year?
- 4 minus 1.
- 3T that will be equal to 2.67.
- Then after 1 meter we have a roller support.
- This is the complete free body diagram of being CD.
- We are done with the free body diagram.

Example or application from the lecture

- So AY is upward, so plus AY don't consider this because this is moment of a couple, it's not a force.
- When you think of AC, consider the reactions.
- Then when you consider this AC separately, you have to consider the reactions at P at C.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 10:00–14:59

Start with the first one sigma fx 0, there is only one horizontal reaction, CS, so CS will be 0.

Key details

- Next I'll go for moment equation because y direction we have two unknowns.
- Then go in this direction towards right.
- The next force we will get is 10 kilometer.
- How is the direction of 10 kilometer?

- Look at this.
- It is clockwise with respect to C.
- Minus forces 10 into distances 1 point three.
- The next forces R D it is look at this the direction.
- Counterclockwise so plus Rd into distances, total distance that is 1 plus 4, 5 meters.
- We have only one unknown that is Rd.
- Simply 5 this you get the value of R d s 2 point double six seven kilo Newton.
- Next, we are left with one more unknown C by.
- Once you apply that, plus C by because upward, downwards, or minus 10, upwards or plus R D will be equals to 0.
- Since we know the value of Rd, substitute here, you will get the value of C by S.
- 7.334 kilogram and our assumption is correct because the answer is positive.
- CY we are taking it as upward.
- The joint C is in equilibrium, so whatever the vertical reaction exerted in this beam C D.
- C D CY is upward.
- That should be balanced by the reaction from the beam AC.
- Then only upward and downward C will get cancelled, the joint C will be in a state of equilibrium.
- Before that we have to convert this UV L into a point, or again we know the formula R is equal to W into L by W L by 2.
- W here is we know from here it is 3 kilon 5 kilonewton.
- Then span is 2 plus 2 that is 4 by 2 you will get the value of 10 kilometer.
- Next way exactly this will be acting with respect to C, with respect to C L by 3, whereas with respect to this point 2 L by 3.
- After that at A we have two reactions, horizontal, vertical.
- Next 3 meters we don't have any loading.
- Then we have this.
- Next we have the resultant of this UV, it is at a distance of L by 3 with respect to C.
- If it is 1.33, then with respect to this point it is going to be 4 minus 1.33 kilonewt.
- Already we know the value of C y here, that is 7.834.
- Three announce ax a y then r b.
- Start with the first condition of equilibrium sigma fx zero.
- Then go in this direction.
- The first course is twenty five.
- Look at the direction up in the right hand thumb rule.
- It is in the clockwise direction.

Example or application from the lecture

- Consider moment about C0.
- If when you consider moment about C0, don't consider C S and C Y because they are acting at the same point, moment will be 0.
- But remember when we consider the beam AC here, from the previous thing we got C by S 1.7334 kN upward.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 15:00–19:59

Just go in this direction.

Key details

- Next course is R B.
- Look at the direction up here right hand rule.
- It is counterclockwise direction.
- Next we have 10 kN.
- Look at the direction.
- It is clockwise.
- That will be equals to from year to year 2.67 plus 3 plus 3.
- That is a perpendicular distance for 10.
- Then look at C y it is clockwise so minus Cy into distances from year to year 4 plus 3 plus 3.
- Is last condition sigma f y 0.
- twelve point nine six kilonewton.
- Remember reaction at A will be equal to the root of A X square plus AY square.
- Since Ax is zero reaction at A is A by the answers are tableture.
- With this, I am going to conclude today's session.
- Go to the assignment questions, then question bank questions.
- If you come across any doubts, you can post your queries in PESO forum.

Example or application from the lecture

- And no need to consider CS because it is Z or it is on the same line of action.
- If you consider sigma f y 0, it is upwards, so plus a y then minus 25, plus b minus 10, then minus c y will be equal to 0.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 41.1 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 41.1 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_41.1_Beams_Numerical__147becf694d21417.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.

5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

C Revision and retrieval practice